Aircraft/Airline Low-Risk Potential Analysis

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Overview

- Identify the low-risk aircraft for airline industry expansion.
- Support informed purchasing decisions for the new division.

Data & Methods

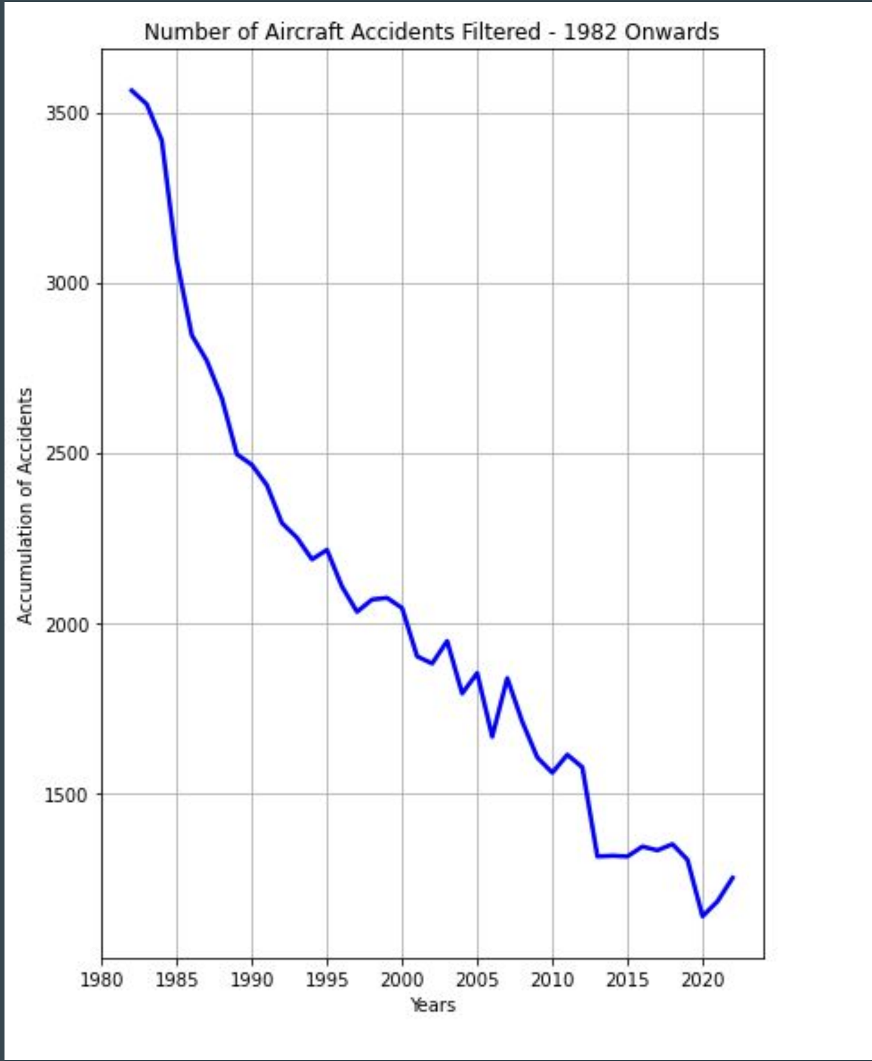
The NTSB aviation accident database will be used for investigation

Fours analysis will be explored

- Aircraft Incident Analysis: Number of aircraft incidents over time.
- Injury Trend Analysis: Fatal, Serious and Minor injuries for the years 1982 thru 2022.
- Top 10 Manufacture Accidents: Analysis of accident occurrences my manufacturer.
- Top 10 Engine Accidents: Analysis of accidents based on engine failures

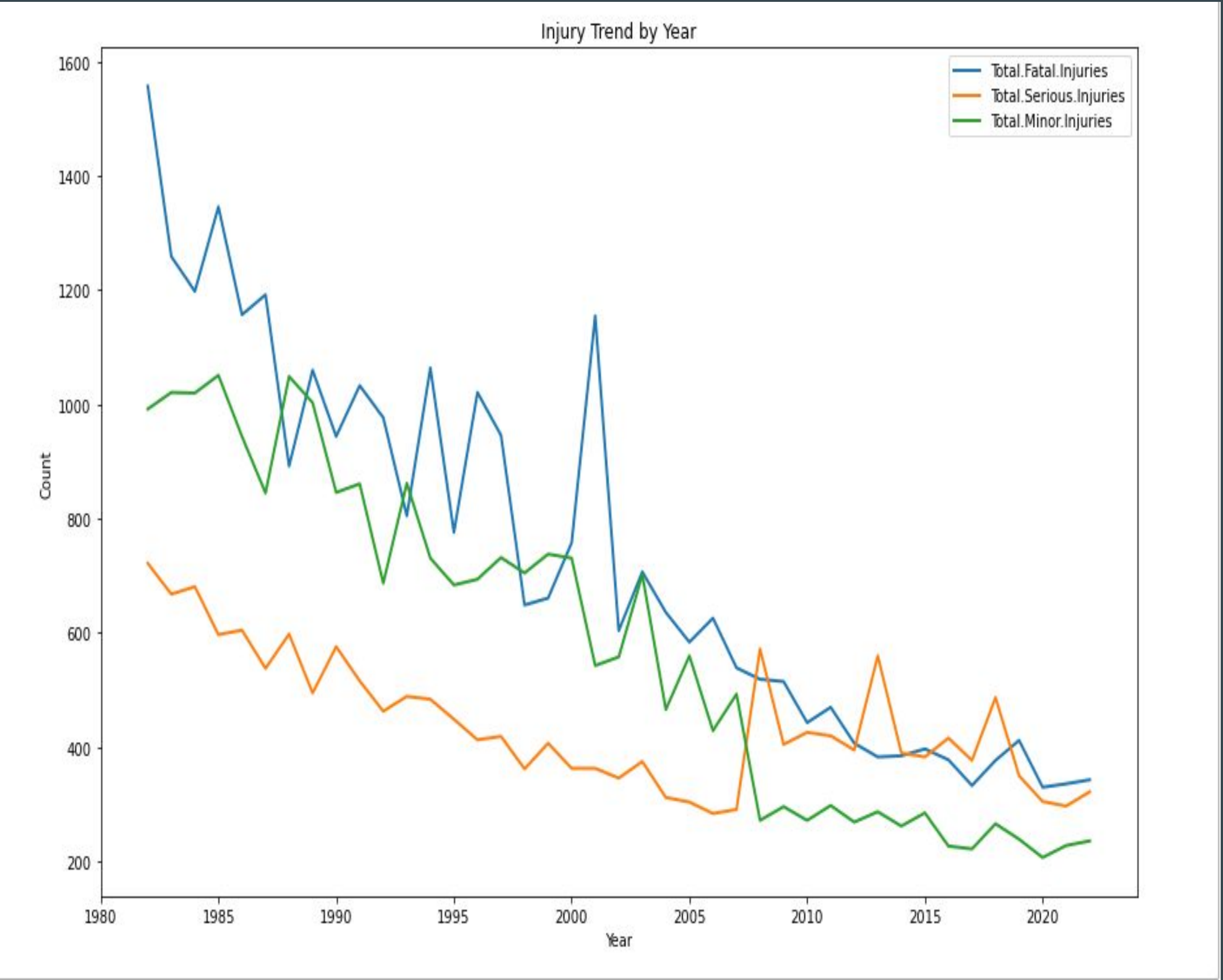
Data Analysis - 1

- Showing downward trend in accidents



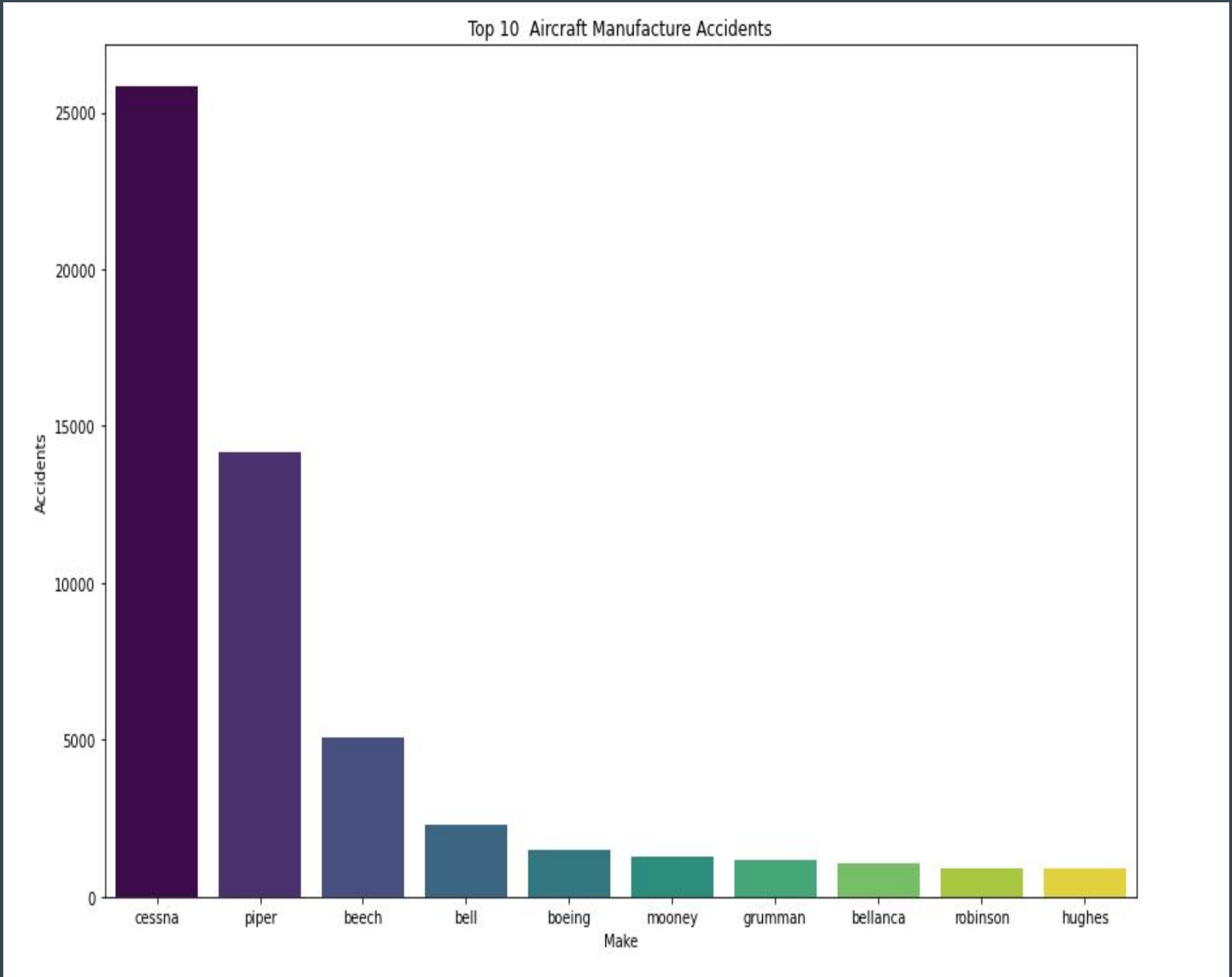
Data Analysis - 2

- Following the downward trend for Fatal, Serious and Minor injuries



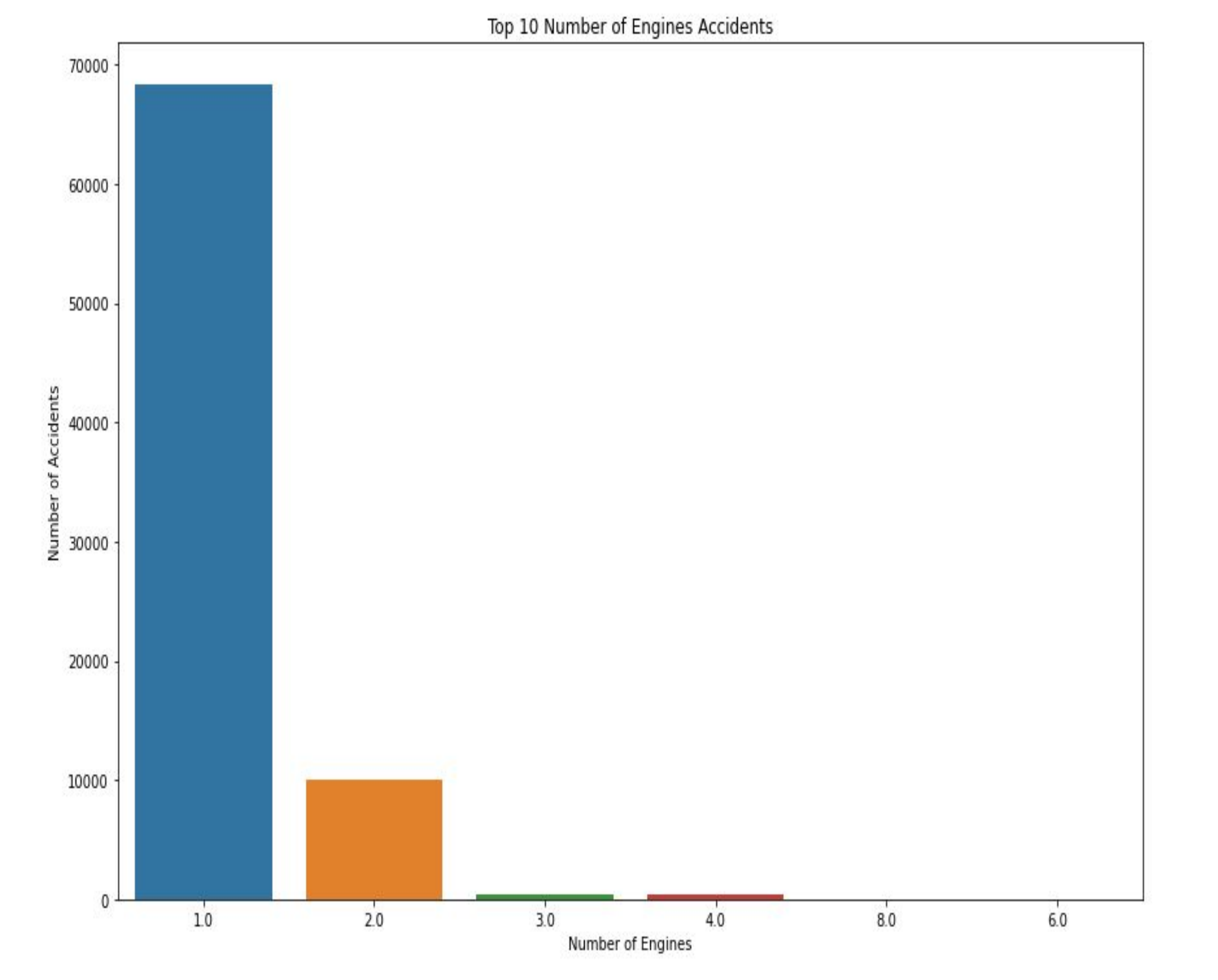
Data Analysis - 3

- Top 10 Aircraft Manufacture accidents



Data Analysis - 4

- Top 10 Number of Engines Accidents



Conclusions

- Significant decline in airline incidents from 1982 to present , showing safety improvements

- Single - engine aircraft seems to have a more accidents than twin engine aircraft
- Two engine aircraft provides good balance of power, efficiency and redundancy.

  Twin engines allow flight to continue after one engine
  fails

- Findings show aviation safety progress and ongoing risks with smaller aircraft

Limitations

- Data mainly represents the U.S., so conclusions focus on U.S. aviation trends

Recommendations

- Expanding data to more countries would improve global insights

Next Steps

- Collect more global data and improve accuracy
- Analyze trends to enhance aviation safety.
- Collaborate with industry stakeholders and regulatory agencies.

Thanks

Tableau link:

https://public.tableau.com/app/profile/joe.mendoza/viz/AircraftAirline_Low-RiskPotential_Analysis_1/Dashboard1

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